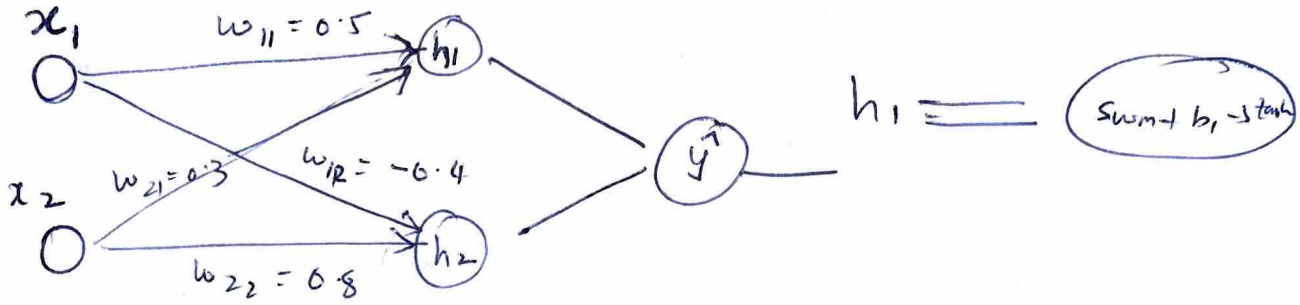
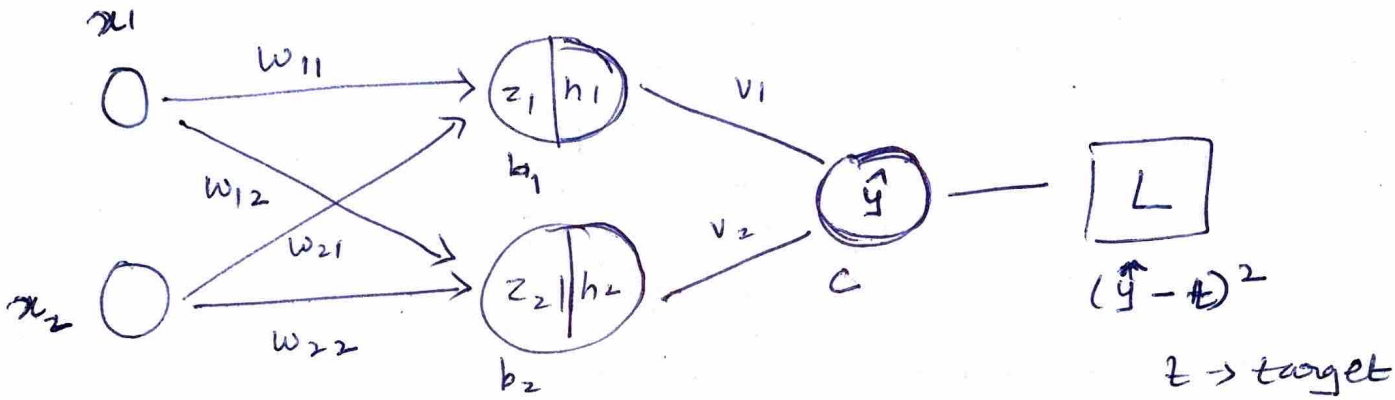
NN





(1) $z_1 = x_1 w_{11} + x_2 w_{21} + b_1$; $h_1 = \tanh(z_1)$

(2) $z_2 = x_1 w_{12} + x_2 w_{22} + b_2$; $h_2 = \tanh(z_2)$

(3) $\hat{y} = v_1 h_1 + v_2 h_2 + c$

(4) $L = (\hat{y} - t)^2$

Q? :- if nudge x_1, how much L moves? w.r.t for x_2.

Forward pass ②

$x_1 = 1$ $w_{11} = 0.5$ $w_{12} = +0.3$ $z_1 = x_1 w_{11} + x_2 w_{21} + b_1$

$x_2 = 0.5$ $w_{21} = -0.4$ $w_{22} = 0.8$

$b_1 = 0.1$ $b_2 = -0.2$

~~$z_1 = 1 \cdot 0.5 + 0.5 \times 0.3 = 0.5$~~

① $z_1 = 1 \times 0.5 + 0.5 \times (0.4) + 0.1$

$= 0.5 - 0.2 + 0.1 = 0.4$

② $h_1 = \tanh(z_1) = 0.380$

③ $z_2 = x_1 w_{12} + x_2 w_{22} + b_2$

$= 1 \cdot 0.3 + 0.5 \times 0.8 + (-0.2)$

$= 0.3 + 0.4 - 0.2 = \underline{0.5}$

④ $h_2 = \tanh(z_2) = 0.462$

⑤ $\hat{y} =$ ✓ p10

(3)

(5) $\hat{y} = h_1 v_1 + h_2 v_2 + c$

$c = 0.2$

$v_1 = 1$

$v_2 = -0.5$

$= 0.380 \times 1 + (0.462) \times (-0.5) + 0.2$

$= 0.38 - 0.231 + 0.2$

$= 0.58 - 0.231 = 0.35$

6

4225

(6) $L = (\hat{y} - t)^2 = (0.35 - 1)^2$

$= 0.65^2 = 0.4225$

Slope of x_1 is if a nudge x_1 by small amt how much does L move?

$\hat{y} = 0.349 \quad L = 0.4224$

$= 0.351 \quad L = 0.4212$

0.0012

$L = (\hat{y} - t)^2$

$\frac{\partial L}{\partial y} = 2(\hat{y} - t)$

$= 2 \times -0.649$

$= -1.298$

④

⑦ $\hat{y}\uparrow \quad L\downarrow$ 반대!

$\hat{y} = v_1 h_1 + v_2 h_2 + c$

if $c \uparrow 0.001$ then $\hat{y}\uparrow 0.001$

$$\frac{\partial L}{\partial \hat{y}} = 2(\hat{y} - t)$$

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①

$$\frac{\partial L}{\partial c} = \frac{\partial \hat{y}}{\partial c} \times \frac{\partial L}{\partial \hat{y}} = 2(\hat{y} - t)\frac{\partial \hat{y}}{\partial c}$$

$$\frac{\partial \hat{y}}{\partial c} = 1 \quad \text{so} \quad \frac{\partial L}{\partial c} = 2(\hat{y} - t)$$

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② $$\frac{\partial L}{\partial v_1} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial v_1}$$

$$= 2(\hat{y} - t) \times h_1$$

③ $$\frac{\partial L}{\partial v_2} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial v_2} = 2(\hat{y} - t) h_2$$

④ $$\frac{\partial L}{\partial h_1} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial h_1} = 2(\hat{y} - t) v_1$$

⑤ $$\frac{\partial L}{\partial h_2} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial h_2} = 2(\hat{y} - t) v_2$$

1) $\frac{\partial L}{\partial \hat{y}} = 2(\hat{y} - t)$

2) $\frac{\partial L}{\partial c} = 2(\hat{y} - t) \times 1$

3) $\frac{\partial L}{\partial v_1} = 2(\hat{y} - t) \times h_1$

4) $\frac{\partial L}{\partial v_2} = 2(\hat{y} - t) \times h_2$

5) $\frac{\partial L}{\partial h_1} = 2(\hat{y} - t)\, v_1$

6) $\frac{\partial L}{\partial h_2} = 2(\hat{y} - t)\, v_2$

7) $\frac{\partial L}{\partial z_1} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial h_1} \times \frac{\partial h_1}{\partial z_1}$

$= \frac{\partial L}{\partial h_1} \times \frac{\partial h_1}{\partial z_1}$

$\frac{\partial h_1}{\partial z_i} = \frac{\partial}{\partial z_i}(\tanh(z_1))$

$\frac{\partial L}{\partial z_1} = 2(\hat{y} - t)\, v_1 \times (1 - \tanh^2(z_1))$

$= 2(\hat{y} - t)\, v_1 (1 - h_1^2)$

8) $\frac{\partial L}{\partial z_2} = \frac{\partial L}{\partial h_2} \times \frac{\partial h_2}{\partial z_2} = 2(\hat{y} - t) v_2 \times (1 - \tanh^2(z_2))$

$= 2(\hat{y} - t) v_2 (1 - h_2^2)$

9) $\frac{\partial L}{\partial w_{11}} = \frac{\partial L}{\partial z_1} \frac{\partial z_1}{\partial w_{11}} = 2(\hat{y} - t)\, v_1 (1 - h_1^2)\, x_1$

L
$\hat{y}$ 1
v_1 2
h_1 3
c 4
v_2 5
h_2 6
z_1 7
z_2 8
x_1 9
x_2 10
w_{11} 11
w_{12} 12
w_{21} 13
w_{22} 14
b_1 15
b_2 16

10) $\frac{\partial L}{\partial w_{21}} = 2(\hat{y} - t) v_1 (1 - h_1^2) x_2$ $\quad \frac{\partial L}{\partial z_1} \times \frac{\partial z_1}{\partial w_{21}}$

11) $\frac{\partial L}{\partial b_1} = 2(\hat{y} - t) v_1 (1 - h_1^2) \times 1$ $\quad \frac{\partial L}{\partial z_1} \times \frac{\partial z_1}{\partial b_1}$

12) $\frac{\partial L}{\partial w_{12}} = \frac{\partial L}{\partial z_2} \frac{\partial z_2}{\partial w_{12}} = 2(\hat{y} - t) v_2 (1 - h_2^2) x_1$

13) $\frac{\partial L}{\partial w_{22}} = 2(\hat{y} - t) v_2 (1 - h_2^2) x_2$ $\quad \frac{\partial L}{\partial z_2} \times \frac{\partial z_2}{\partial w_{22}}$

14) $\frac{\partial L}{\partial b_2} = 2(\hat{y} - t) v_2 (1 - h_2^2) \times 1$ $\quad \frac{\partial L}{\partial z_2} \times \frac{\partial z_2}{\partial b_2}$

$\hat{y} = 0.349 \quad h_1 = 0.380 \quad h_2 = 0.462 \quad x_1 = 1 \quad x_2 = 0.5$

$v_1 = 1.0 \quad v_2 = -0.5 \quad t = 1$

1) $2(\hat{y} - t) = 2(0.349 - 1) = \underline{-1.302}$

2) $\frac{\partial L}{\partial z_1} = -1.302 \times 1.0 \times (1 - 0.38^2) = \underline{-1.1139}$

3) $\frac{\partial L}{\partial z_2} = -1.302 \times -0.5 \times (1 - 0.462^2)$

$= \underline{0.512}$

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1) $\frac{\partial L}{\partial \hat{y}} = -1.302$

2) $\frac{\partial L}{\partial c} = -1.302$

3) $\frac{\partial L}{\partial v_1} = -1.302 \times 0.38 = -0.495$

4) $\frac{\partial L}{\partial v_2} = -1.302 \times 0.462 = -0.601$

5) $\frac{\partial L}{\partial h_1} = -1.302 \times 1 = -1.302$

6) $\frac{\partial L}{\partial h_2} = -1.302 \times -0.5 = 0.651$

7) $\frac{\partial L}{\partial z_1} = -1.1139$

8) $\frac{\partial L}{\partial z_2} = 0.512$

9) $\frac{\partial L}{\partial w_{11}} = -1.1139 \times 1$

$= -1.1139$

10) $\frac{\partial L}{\partial w_{21}} = -1.1139 \times 0.5$

$= -0.556$

11) $\frac{\partial L}{\partial b_1} = -1.1139$

12) $\frac{\partial L}{\partial w_{12}} = \underline{0.512}$

13) $\frac{\partial L}{\partial w_{22}} = 0.512 \times 0.5$

$= 0.256$

14) $\frac{\partial L}{\partial b_2} = 0.512$